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Rotary electrical plug

Abstract

A rotary electrical plug is provided. The rotary electrical plug includes a rotary base, a first insert, a second insert, a first conductive connection lug, and a second conductive connection lug. The first insert and the second insert are circumferentially limited and mounted on the rotary base, the first insert is provided with a first conductive circular portion, the second conductive connection lug is provided with a second conductive circular portion, and the first conductive circular portion, the second conductive circular portion, and the rotary base are coaxial with each other. The first conductive connection lug is electrically connected to the first insert via the first conductive circular portion, and the second conductive connection lug is electrically connected to the second insert via the second conductive circular portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese patent application No. 202320165241.6, filed on Jan. 16, 2023, the content of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of an electrical plug and an electrical socket, and in particular, to a rotary electrical plug.

BACKGROUND

(3) An electrical plug is usually matched with an electrical socket, which enables an electronic product with the electrical plug to be energized.

(4) At present, an insert of a conventional plug and a power cord are not rotatable. The electrical socket is usually mounted to the wall, and a location of a jack of the electrical socket is also fixed. However, in an actual application, a direction of the power cord may need to change in an external environment. The conventional electrical plug cannot change the direction of the power cord. Moreover, a thickness of the conventional electrical plug is large and extends outwardly from the wall, thereby preventing cabinets and other furniture from being placed in front of the plug.

SUMMARY

(5) The present disclosure provides a rotary electrical plug to solve the above-mentioned problem.

(6) The rotary electrical plug includes a rotary base, a first insert, a second insert, a first conductive connection lug, and a second conductive connection lug. The first insert and the second insert are circumferentially limited and mounted on the rotary base, the first insert is provided with a first conductive circular portion, the second conductive connection lug is provided with a second conductive circular portion, and the first conductive circular portion, the second conductive circular portion, and the rotary base are coaxial with each other.

(7) The first conductive connection lug is electrically connected to the first insert via the first

conductive circular portion, and the second conductive connection lug is electrically connected to the second insert via the second conductive circular portion.

(8) In some embodiments, the rotary base is provided with a first groove, and the first conductive circular portion is disposed in the first groove and attached to a bottom surface of the first groove. The first insert is provided with a main portion, and the main portion of the first insert is vertically connected to the first conductive circular portion and penetrates through the rotary base.

(9) In some embodiments, the second conductive circular portion is rotatably mounted to a periphery of the rotary base. A contact connection portion is defined by the second insert extending towards the second conductive circular portion, and the second insert is in contact with and cooperated with the second conductive circular portion via the contact connection portion.

(10) In some embodiments, the rotary electrical plug further includes an inner frame. The rotary base is rotatably mounted on the inner frame. The inner frame abuts against the second conductive circular portion to support the second conductive connection lug.

(11) In some embodiments, the rotary electrical plug further includes a collision member. The collision member is circumferentially limited and mounted on the rotary base and facing the inner frame, to control a collision between the collision member and the inner frame driven by the rotary base and produce a sound.

(12) In some embodiments, the collision member includes an elastic body and a collision protrusion, and the collision protrusion is mounted on the elastic body. The inner frame is provided with convex ribs separated from each other sequentially on an inner circumferential wall thereof, a position interference portion is located between the convex ribs and the collision protrusion, and the elastic body is configured to elastically deform relative to the inner frame in response to pushing of the position interference portion, so as to enable the rotary base to drive the collision member to rotate relative to the inner frame.

(13) In some embodiments, the rotary electrical plug further includes a shell, both the inner frame and the rotary base are mounted in the shell. The inner frame is provided with a rotary limiting portion, the shell is provided with a coordinating limiting portion, and the coordinating limiting portion of the shell is coordinated with the rotary limiting portion of the inner frame, so as to circumferentially limit the inner frame to the shell.

(14) In some embodiments, the rotary electrical plug further includes a grounding pin and a third conductive connection lug. The grounding pin is circumferentially limited and mounted on the rotary base, the grounding pin is provided with a circular conductive portion, and a line perpendicular to a surface of the circular conductive portion and an axis of the rotary base are coaxial. The third conductive connection lug is electrically connected to the grounding pin via the circular conductive portion.

(15) In some embodiments, the rotary base is further provided with a second groove and a center hole, and the center hole is in communication with the second groove, where the circular conductive portion is disposed; and the third conductive connection lug is capable of passing through the center hole and contacting and cooperating with the circular conductive portion.

(16) In some embodiments, the rotary base is further provided with an extending convex column, which is disposed in an area where the second groove is located, and the extending convex column is inserted in the grounding pin, so as to circumferentially limit the grounding pin to the rotary base.

(17) The details of one or more embodiments of the present disclosure are set forth in the accompanying drawings and the description below. Other features, objects and advantages of the present disclosure will become apparent from the description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to illustrate the embodiments of the present disclosure more clearly, the drawings used in the embodiments will be described briefly. Apparently, the following described drawings are merely for the embodiments of the present disclosure, and other drawings can be derived by those of ordinary skill in the art without any creative effort.
- (2) FIG. 1 is a schematic diagram of a rotary electrical plug in an embodiment of the present disclosure.
- (3) FIG. 2 is an exploded diagram of a rotary electrical plug in an embodiment of the present disclosure.
- (4) FIG. 3 is a schematic diagram of a partial structure of a rotary electrical plug in an embodiment of the present disclosure.
- (5) FIG. 4 is a schematic diagram of the partial structure of the rotary electrical plug in another view in an embodiment of the present disclosure.
- (6) FIG. 5 is a schematic diagram of a rotary base in an embodiment of the present disclosure.
- (7) FIG. 6 is a schematic diagram of the rotary base in another view in an embodiment of the present disclosure.
- (8) FIG. 7 is a schematic diagram of a first insert in an embodiment of the present disclosure.
- (9) FIG. 8 is a schematic diagram of a second insert in an embodiment of the present disclosure.
- (10) FIG. 9 is a schematic diagram of a second conductive connection lug in an embodiment of the present disclosure.
- (11) FIG. 10 is a schematic diagram of an inner frame in an embodiment of the present disclosure.
- (12) FIG. 11 is a schematic diagram of a collision member in an embodiment of the present disclosure.
- (13) FIG. 12 is a schematic diagram of a lower cover in an embodiment of the present disclosure.
- (14) FIG. 13 is a schematic diagram of a grounding pin in an embodiment of the present disclosure.
- (15) In the figures, **100** represents a rotary electrical plug, **10** represents a rotary base, **11** represents a first groove, **111** represents a bottom surface of the first groove, **12** represents a limiting groove, **121** represents a groove wall, **13** represents a blocking protrusion, **14** represents a gap, **15** represents a second groove, **16** represents a center hole, **17** represents an extending convex column, **20** represents a first insert, **21** represents a first conductive circular portion, **22** represents a main portion of the first insert, **30** represents a second insert, **31** represents an insert connection portion, **40** represents a first conductive connection lug, **50** represents a second conductive connection lug, **51** represents a second conductive circular portion, **60** represents an inner frame, **61** represents a first slot, **62** represents a convex rib, **63** represents a rotary limiting portion, **70** represents a collision member, **71** represents an elastic body, **72** represents a collision protrusion, **80** represents a shell, **81** represents a lower cover, **811** represents a coordinating limiting portion, **82** represents an upper cover, **83** represents a screw, **91** represents a grounding pin, **911** represents a circular conductive portion, **92** represents a third conductive connection lug, **101** represents a power line, **1011** represents a net tail and rubber coating portion, and **102** represents a rotary carrier.

DETAILED DESCRIPTION

- (16) The present disclosure will now be described in detail with reference to the accompanying drawings and embodiments in order to make the objects, technical solutions, and advantages of the present disclosure clearer. It should be understood that the specific embodiments described herein are only for explaining the present disclosure, and not intended to limit the present disclosure.
- (17) It should be noted that when an element is considered to be “mounted on” another element, it can be directly on another element or there may be a centered element. When an element is considered to be “disposed on” another element, it can be disposed directly on another element or there may be a centered element at the same time. When an element is considered to be “fixed to” another element, it can be directly fixed to another element, or there may be a centered element.

(18) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as a skilled person in the art would understand. The terms used in the specification of the present disclosure are only for the purpose of describing specific embodiments and are not intended to limit the present disclosure. The term “and/or” as used herein may include any and all combinations of one or more of the relevant listed items.

(19) A rotary electrical plug **100** provided in the present disclosure refers to a plug that complies with the American National Standard specifically.

(20) Referring to FIG. 2, FIG. 7, and FIG. 9, the rotary electrical plug **100** provided in an embodiment of the present disclosure includes a rotary base **10**, a first insert **20**, a second insert **30**, a first conductive connection lug **40**, and a second conductive connection lug **50**. The first insert **20** and the second insert **30** are circumferentially limited and mounted on the rotary base **10**, the first insert **20** is provided with a first conductive circular portion **21**, the second conductive connection lug **50** is provided with a second conductive circular portion **51**, and the first conductive circular portion **21**, the second conductive circular portion **51**, and the rotary base **10** are coaxial with each other. The first conductive connection lug **40** is corresponding to the first insert **20**, the first conductive connection lug **40** is electrically connected to the first insert **20** via the first conductive circular portion **21**, the second conductive connection lug **50** is corresponding to the second insert **30**, and the second conductive connection lug **50** is electrically connected to the second insert **30** via the second conductive circular portion **51**.

(21) That is, the rotary electrical plug **100** may apply structural features of the first conductive circular portion **21** and the second conductive circular part **51**, so that the rotary electrical plug **100** may ensure a rotation of the first insert **20** and the second insert **30** relative to the rotary electrical plug **100** on the basis of electrical conductivity of the first insert **20** and the second insert **30**. A direction of a power line **101** of the rotary electrical plug **100** may be changed to mate the rotary electrical plug **100** with the electrical socket mating. Meanwhile, the first conductive circular portion **21** may be integrated in the first insert **20**, and the second conductive circular portion **51** may be integrated in the second conductive connection lug **50**, so as to facilitate arrangement of the first insert **20**, the second insert **30**, the first conductive connection lug **40** and the second conductive connection lug **50** in the rotary electrical plug **100**. Internal space of the rotary electrical plug **100** may be made full use, which plays a role in reducing an overall thickness of the rotary electrical plug **100**.

(22) It should be noted that the first insert **20** may be a zero-wire insert, the second insert **30** may be a fire-wire insert specifically. For one skilled in the art, the first insert **20** may be the fire-wire insert, the second insert **30** may be the zero-wire insert.

(23) Referring to FIG. 6 and FIG. 7, in an embodiment, the rotary base **10** may be provided with a first groove **11**, and the first conductive circular portion **21** may be disposed in the first groove **11** and attached to a bottom surface **111** of the first groove **11**. The first insert **20** may be provided with a main portion **22**, and the main portion **22** of the first insert **20** may be vertically connected to the first conductive circular portion **21** and penetrate through the rotary base **10**. This specifically may enable assembly of the first insert **20** on the rotary base **10**, and the assembly of the first insert **20** on the rotary base **10** may be limited by the fit between the first conductive circular portion **21** and the bottom surface **111** of the first groove **11**, and the main portion **22** of the first insert **20** penetrating through the rotating base **10**.

(24) The first groove **11** may match the first conductive circular portion **21**. Specifically, the first groove **11** may be a circular groove, so as to reduce an area of the first groove **11**, and thus play a role in reducing a size of the rotary base **10**. The first groove **11** may also be a groove in other shapes, a diameter of which is greater than that of the first conductive circular portion **21**, such as an avoiding oval shape, or an irregular shape.

(25) Referring to FIG. 3 and FIG. 4, in an embodiment, the second conductive circular portion **51** may be rotatably mounted to a periphery of the rotary base. A contact connection portion **31** may be

defined by the second insert **30** extending towards the second conductive circular portion **51**, and the second insert **30** is in contact with and cooperated with the second conductive circular portion **51** via the contact connection portion **31**. In this way, an assembly of the second conductive connection lug **50** and the second insert **30** on the rotary base **10** may be realized.

(26) Referring to FIG. **8**, the number of the contact connection portion **31** may be more than one, such as two. Two contact connection portions **31** are capable of abutting against the rotary base **10** and in contact with the second conductive circular portion **51**, respectively, so as to ensure an electrical connection between the second insert **30** and the second conductive connection lug **50**.

(27) Referring to FIG. **5**, the rotary base **10** may be further provided with a limiting groove **12** and a blocking protrusion **13** matched with the second insert **30**. The second insert **30** may be partially inserted into the limiting groove **12**, and abut against the blocking protrusion **13**. The circumferential and restrictive mounting of the second insert **30** on the rotary base **10** may be realized by the abutment among the second insert **30**, the blocking protrusion **13**, and a groove wall **121** of the limiting groove **12**.

(28) Referring to FIG. **2**, the rotary electrical plug **100** may further include an inner frame **60**. The rotary base **10** is rotatably mounted on the inner frame **60**. The inner frame **60** may abut against the second conductive circular portion **51** to support the second conductive connection lug **50**. In other words, the inner frame **60** is configured to support the rotary **10** and the second conductive connection lug **50** in the rotary electrical plug **100**, resulting in realizing assembly of the rotary base **10** and the second conductive connection lug **50** in the rotary electrical plug **100**.

(29) Referring to FIG. **10**, the inner frame **60** may be provided with a first slot **61**, and the first conductive connection lug **40** is capable of penetrating through the first slot **61** and in contact with the first conductive circular portion **21** on the first insert **20**. The number of the first conductive connection lug **40** in contact with the first conductive circular portion **21** corresponds to the number of the first slot **61**, and a width of the first slot **61** may be less than a width of the first conductive circular portion **21**, to facilitate limiting the first conductive circular part **21** on the rotary base **10** by the inner frame **60**.

(30) Referring to FIG. **2**, in an embodiment, the rotary electrical plug **100** may further include a collision member **70**. The collision member **70** may be circumferentially limited and mounted on the rotary base **10** and facing the inner frame **60**. The collision member **70** is configured to control a collision between the collision member **70** and the inner frame **60** driven by the rotary base **10** and generate a sound. In other words, the rotary electrical plug **100** in the rotation may accompany with the sound, so as to improve an intuitive feeling of the user when using the rotating plug **100**.

(31) Referring to FIG. **10** and FIG. **11**, the collision member **70** may include an elastic body **71** and a collision protrusion **72**, and the collision protrusion **72** may be mounted on the elastic body **71**. The inner frame **60** may be provided with convex ribs **62** separated from each other sequentially on an inner circumferential wall thereof, a position interference portion may be located between the convex ribs **62** and the collision protrusion **72**, and the elastic body **71** is configured to elastically deform relative to the inner frame in response to pushing of the position interference portion, so as to enable the rotary base **10** to drive the collision member **70** to rotate relative to the inner frame **60**. In other words, the collision member **70** is capable of generating a sound by a collision between the collision protrusion **72** and the convex ribs **62**. The elastic body **71** is capable of elastically deforming driven by the collision protrusion **72**, so that the collision protrusion **72** may smoothly overcome the blockage of the convex ribs **62**. For every rotation of the rotary base **10**, the collision protrusion **72** of the collision member **70** may collide with the convex ribs **62** of the inner frame **60** and generate the sound. Specifically, a clattering sound similar to that of a mechanical watch. It should be noted that the collision member **70** may be a resilient metal sheet. In order to realize the assembly of the collision member **70** on the rotary base **10**, the rotary base **10** may be provided with a gap **14** matched with the collision member **70**, so that the rotary base **10** may drive the collision member **70** to rotate relative to the inner frame **60**.

(32) Referring to FIG. 10 and FIG. 12, the rotary electrical plug 100 may further include a shell 80, both the inner frame 60 and the rotary base 10 may be mounted in the shell 80. The inner frame 60 may be provided with a rotary limiting portion 63, the shell 80 may be provided with a coordinating limiting portion 811, and the coordinating limiting portion 811 of the shell 80 may be coordinated with the rotary limiting portion 63 of the inner frame 60, so as to circumferentially limit the inner frame 60 to the shell 80. In this way, the use requirement of assembly of the inner frame 60 in the rotary electrical plug 100 may be realized.

(33) Referring to FIG. 1, the shell 80 may include a lower cover 81 and an upper cover 82. The upper cover 82 may be mounted to the lower cover 81 by means of clamping, and then fixed with a screw 83. The coordinating limiting portion 811 may be specifically located on the lower cover 81.

(34) It should be noted that the coordination between the coordinating limiting portion 811 and the rotary limiting portion 63 may be specifically a concave-convex coordination. The number of the rotary limiting portion 63 may be two, and two rotary limiting portions 63 may be located at both sides of the inner frame 60, which will not be elaborated herein.

(35) Referring to FIG. 2 and FIG. 13, the rotary electrical plug 100 may further include a grounding pin 91 and a third conductive connection lug 92. The grounding pin 91 may be circumferentially limited and mounted on the rotary base 10. The grounding pin 91 may be provided with a circular conductive portion 911, and a line perpendicular to a surface of the circular conductive portion 911 and an axis of the rotary base 10 are coaxial. The third conductive connection lug 92 may be corresponding to the grounding pin 91, and the third conductive connection lug 92 may be electrically connected to the grounding pin 91 via the circular conductive portion 911. In this way, the rotary electrical plug 100 may have a grounding function, and the circular conductive portion 911 of the grounding pin 91 may be in contact with the third conductive connection lug 92 to meet the requirement of rotation of the rotary electrical plug 100 as a whole.

(36) It should be noted that all of the grounding pin 91, the third conductive connection lug 92, the first insert 20, the first conductive connection lug 40, the second insert 30, and the second conductive connection lug 50 may be bent by hardware copper one-piece molding, with greater strength than that of a conventional structure, better conductivity and contact, and less resistance.

(37) Referring to FIG. 5, in an embodiment, the rotary base 10 may further include provided with a second groove 15 and a center hole 16, and the center hole 16 may be in communication with the second groove 15, where the circular conductive portion 911 is disposed. The third conductive connection lug 92 is capable of passing through the center hole 16 and contacting and cooperating with the circular conductive portion 911. In this way, the grounding pin 91 may be connected to the rotary base 10, so that the grounding pin 91 may be assembled on the rotary base 10 in a limited position.

(38) The rotary base 10 may be further provided with an extending convex column 17, which may be disposed in an area where the second groove 15 is located, and the extending convex column 17 may be inserted in the grounding pin 91, so as to circumferentially limit the grounding pin 91 to the rotary base 10. In this way, circumferential restriction of the grounding pin 91 on the rotary base 10 may be realized.

(39) Referring to FIG. 1, it should be noted that the rotary electrical plug 100 may further include a rotary carrier 102. The rotary carrier 102 may cover on the rotary base 10, and the first insert 20, the second insert 30, and grounding pin 91 on the rotary base 10 can penetrate through the rotary carrier 102 and extend out of the shell 80. The power line 101 of the rotary electrical plug 100 extending out of the shell 80 may be provided with a net tail and rubber coating portion 1011 in the injection molding, and the net tail and rubber coating portion 1011 may abut against the shell 80, so that the abutting cooperation between the net tail and rubber coating portion 1011 and the shell 80 may enable the rotary electrical plug 100 to successfully pass a swing test.

(40) In summary, the rotary electrical plug 100 provided in the present disclosure may achieve a rotary function. The direction of the power line 101 of the rotary electrical plug 100 may be

changed to mate the rotary electrical plug **100** with the electrical socket mating. Meanwhile, the rotary electrical plug **100** may make full use of the internal space of the rotary electrical plug **100**, which plays a role in reducing the overall thickness of the rotary electrical plug **100**.

(41) The technical features of the above-described embodiments may be combined in any combination. For the sake of brevity of description, not all possible combinations of the technical features in the above embodiments are described. However, as long as there is no contradiction between the combinations of these technical features, all should be considered as within the scope of this disclosure.

(42) The above-described embodiments are merely illustrative of several embodiments of the present disclosure, and the description thereof is relatively specific and detailed, but is not to be construed as limiting the scope of the disclosure. It should be noted that a number of variations and modifications may be made by those skilled in the art without departing from the spirit and scope of the disclosure. Therefore, the scope of the disclosure should be determined by the appended claims.

Claims

1. A rotary electrical plug, comprising a rotary base, a first insert, a second insert, a first conductive connection lug, and a second conductive connection lug, wherein the first insert and the second insert are circumferentially limited and mounted on the rotary base, the first insert is provided with a first conductive circular portion, the second conductive connection lug is provided with a second conductive circular portion, and the first conductive circular portion, the second conductive circular portion, and the rotary base are coaxial with each other; and the first conductive connection lug is electrically connected to the first insert via the first conductive circular portion, and the second conductive connection lug is electrically connected to the second insert via the second conductive circular portion.
2. The rotary electrical plug of claim 1, wherein the rotary base is provided with a first groove, and the first conductive circular portion is disposed in the first groove and attached to a bottom surface of the first groove; wherein the first insert is provided with a main portion, and the main portion of the first insert is vertically connected to the first conductive circular portion and penetrates through the rotary base.
3. The rotary electrical plug of claim 1, wherein the second conductive circular portion is rotatably mounted to a periphery of the rotary base; and a contact connection portion is defined by the second insert extending towards the second conductive circular portion, and the second insert is in contact with and cooperated with the second conductive circular portion via the contact connection portion.
4. The rotary electrical plug of claim 1, further comprising an inner frame, wherein the rotary base is rotatably mounted on the inner frame, and the inner frame abuts against the second conductive circular portion to support the second conductive connection lug.
5. The rotary electrical plug of claim 4, further comprising a collision member, wherein the collision member is circumferentially limited and mounted on the rotary base and facing the inner frame, to control a collision between the collision member and the inner frame driven by the rotary base and produce a sound.
6. The rotary electrical plug of claim 5, wherein the collision member comprises an elastic body and a collision protrusion, and the collision protrusion is mounted on the elastic body; and the inner frame is provided with convex ribs separated from each other sequentially on an inner circumferential wall thereof, a position interference portion is located between the convex ribs and the collision protrusion, and the elastic body is configured to elastically deform relative to the inner frame in response to pushing of the position interference portion, so as to enable the rotary base to drive the collision member to rotate relative to the inner frame.

7. The rotary electrical plug of claim 4, further comprising a shell, wherein both the inner frame and the rotary base are mounted in the shell; and the inner frame is provided with a rotary limiting portion, the shell is provided with a coordinating limiting portion, and the coordinating limiting portion of the shell is coordinated with the rotary limiting portion of the inner frame, so as to circumferentially limit the inner frame to the shell.
8. The rotary electrical plug of claim 1, further comprising a grounding pin and a third conductive connection lug, wherein the grounding pin is circumferentially limited and mounted on the rotary base, the grounding pin is provided with a circular conductive portion, and a line perpendicular to a surface of the circular conductive portion and an axis of the rotary base are coaxial; and the third conductive connection lug is electrically connected to the grounding pin via the circular conductive portion.
9. The rotary electrical plug of claim 8, wherein the rotary base is further provided with a second groove and a center hole, and the center hole is in communication with the second groove, where the circular conductive portion is disposed; and the third conductive connection lug is capable of passing through the center hole and contacting and cooperating with the circular conductive portion.
10. The rotary electrical plug of claim 9, wherein the rotary base is further provided with an extending convex column, which is disposed in an area where the second groove is located, and the extending convex column is inserted in the grounding pin, so as to circumferentially limit the grounding pin to the rotary base.
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